

Bill Of Materials for: TEAM 4 - ECU
Last modified: 5/19/2026
BOM revision: v1.8.1

Reference	Qty	Mfr	Value	Description	Footprint	Mfr Part Number	Dist	Dist Part Number	Cost Ea.	Cost Total	Comments	
A1	☒	1	Adafruit	Adafruit_Feather_HUZZAH32_ESP32	Microcontroller module with ESP32 MCU	Module:Adafruit_Feather	3591	Digikey	1528-2514-ND	\$20.95	\$20.95	
C1,C6	☐	2	KEMET	100u	Polarized capacitor, small US symbol	Capacitor_SMD:CP_Elec_6.3x7.7	A768EB107M1ELAS036	Digikey	399-A768EB107M1ELAS036CT-ND	\$0.55	\$1.10	
C2,C3,C4,C5,C7	☐	5	KYOCERA AVX	0.1u	Unpolarized capacitor, small symbol	Capacitor_SMD:C_1206_3216Metric_Pad1.33x1.80mm_HandSolder	KAF31BR71H104KT	Mouser	581-KAF31BR71H104KT	\$0.29	\$1.45	
D1	☐	1	American Bright LED	YELLOW_LED	Light emitting diode	LED_SMD:LED_1206_3216Metric_Pad1.42x1.75mm_HandSolder	BL-HKB33Q-AV-TR8	Mouser	442-BLHKB33QAVT8	\$0.23	\$0.23	
D2,D3,D5	☐	3	Kingbright	GREEN_LED	Light emitting diode	LED_SMD:LED_1206_3216Metric_Pad1.42x1.75mm_HandSolder	APT3216MGC	Mouser	604-APT3216MGC	\$0.60	\$1.80	
D4,D6,D7	☐	3	onsemi	SS14	40V 1A Schottky Diode, SMA	Diode_SMD:D_SMA	SS14	Mouser	SS14CT-ND	\$0.44	\$1.32	
D9,D10,D11	☐	3	Inlux	Blue_LED_1,2,3	Light emitting diode	LED_SMD:LED_1206_3216Metric_Pad1.42x1.75mm_HandSolder	IN-PI5126ATP8	Mouser	743-IN-PI5126ATP8	\$0.51	\$1.53	
J1	☐	1	GCT	Jack-DC	DC Barrel Jack	Connector_BarrelJack:BarrelJack_GCT_DCJ200-10-A_Horizontal	DCJ200-10-A-K1-K	Digikey	2073-DCJ200-10-A-K1-K-ND	\$0.74	\$0.74	
J2,J3,J4	☐	3	Würth Elektronik	Fan 1,2,3 terminals	Generic screw terminal, single row, 01x02	TerminalBlock_Phoenix:TerminalBlock_Phoenix_MKDS-1.5-2-5.08_1x02_P5.08mm_Horizontal	691137710002	Digikey	732-10955-ND	\$0.41	\$1.23	
J6	☐	1	Sullins Connector Solutions	Conn_02x04_Counter_Clockwise	Generic connector, double row, 02x04,	Connector_PinSocket_2.54mm:PinSocket_2x04_P2.54mm_Vertical	PPTC042LFBN-RC	Digikey	S7072-ND	\$0.64	\$0.64	
Q1,Q2,Q3	☒	3	onsemi	BSS138	50V Vds, 0.22A Id, N-Channel MOSFET, SOT-23	Package_TO_SOT_SMD:SOT-23_Handsoldering	BSS138	Digikey	BSS138CT-ND	\$0.31	\$0.93	Don't buy. David has a lot
R1,R2,R3,R5	☐	4	YAGEO	5.1k	Resistor, small US symbol	Resistor_SMD:R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	RC1206FR-075K1L	Digikey	311-5.10KFRCT-ND	\$0.10	\$0.40	
R6,R7,R8	☐	3	YAGEO	10k	Resistor, small US symbol	Resistor_SMD:R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	RT1206BRD0710KL	Digikey	YAG2329CT-ND	\$0.21	\$0.63	
R9,R10,R11	☐	3	YAGEO	100	Resistor, small US symbol	Resistor_SMD:R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	RC1206FR-07100RL	Digikey	311-100FRCT-ND	\$0.10	\$0.30	
R12,R13,R14	☐	3	YAGEO	1k	Resistor, small US symbol	Resistor_SMD:R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	RC1206FR-071KL	Digikey	311-1.00KFRCT-ND	\$0.10	\$0.30	
R15,R16	☐	2	YAGEO	4.7k	Resistor, small US symbol	Resistor_SMD:R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	RC1206FR-074K7L	Digikey	311-4.70KFRCT-ND	\$0.10	\$0.20	
R17	☐	1	YAGEO	0	Resistor, small US symbol	Resistor_SMD:R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	RC1206JR-070RL	Digikey	311-0.0FRCT-ND	\$0.10	\$0.10	
SW1,SW2	☒	2	VEUXINGA	SW_SPDT	Switch, single pole double throw	switch_DM:switch_dm	s512944	Amazon	s512944	\$0.12	\$0.24	Don't buy. Switch kit David has a few
SW3,SW4	☐	2	Same Sky	SW_Push	Push button switch, generic, two pins	Button_Switch_THT:SW_PUSH_6mm	TS02-66-70-BK-100-LCR-D	Digikey	2223-TS02-66-70-BK-100-LCR-D-ND	\$0.11	\$0.22	
TP1,TP2,TP3,TP4,TP5	☐	5	Keystone Electronics	TestPoint	test point	Connector_Pin-Pin_D0.7mm_L6.5mm_W1.8mm_Flatfork	S011	Digikey	36-S011-ND	\$0.28	\$1.40	
U3	☐	1	Adafruit	SCD30	CO2 Sensor	SCD30:MODULE_SCD30	SCD30	Adafruit	SCD30	\$58.95	\$58.95	
U4	☒	1	Euogeudel	Mini360 12V to 3.3V BUCK	12 V to 3.3 V buck converter	Library:Mini360	MINI360	Amazon	MINI360	\$1.74	\$1.74	Don't Buy. Pack of 5. Bulk order David has a few
U2	☐	1	Adafruit	BME688	Temp/humidity/pressure & gas Sensor	BME680:PS0N80P300X300X100-8N	BME680	Adafruit	BME680	\$19.95	\$19.95	
U1	☐	1	Adafruit	SGP30	TVOC & NOx Sensor	SGP30:SGP30	SGP30	Adafruit	SGP30	\$17.50	\$17.50	
-	☐	1	WINSINN	Fan	WINSINN DC 40mm Fan 12V 4028	-	F402812V	Amazon	747302188349	\$10.99	\$10.99	Buy Pack of 4.
TOTAL:										\$144.84	* This BOM is for one single Board	

VERSION INFO	Comments
Rev	
1.8	5/11/2026
1.8	5/19/2026

Links

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<https://www.digikey.com/en/products/detail/vageo/BC1206FR-075K1L/728942>

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